Github link: https://github.com/ExplosiveSalad/Credit-Card-Skimmers

Task 1: Problem Definition and Dataset Selection

Problem Statement:

Credit card fraud to this day presents a persistent and costly issue for financial systems around the world. As digital and card not present transitions (online shopping, phone orders, subscription payments) have become the norm, chances for fraudulent activity has grown alongside them, and traditional rule-based detection systems (criteria list, blacklists, human review) struggle to keep up with both the ever-growing number of transactions and the constant adaptation of hackers. This project addresses a binary classification problem: given a set of transaction-level features, predict whether an individual credit card transaction is fraudulent or legitimate. The goal is to demonstrate how a data-driven, machine learning-based approach can support automated, scalable fraud screening.

Motivation and Background:

Financial systems process millions of transactions on the daily, making manual review of every transaction time consuming and prone to human error. Machine learning offers a way to automatically flag suspicious transactions in sub real time by learning patterns from data rather than relying on static rules. However, fraud detection poses a difficult modelling challenge: fraudulent transactions are extremely rare compared to legitimate ones, meaning the dataset is highly imbalanced, and the cost of different types of errors is uneven. A missed fraudulent transaction (false negative) results in financial lost, whilst an incorrectly flagged legitimate transaction (false positive) inconveniences a customer and can harm the customer-company trust. This makes the problem a strong contender for exploring not just model accuracy, but metrics that keep up with real-world costs, such as precision, recall, the trade off, and the constant attempt to keep up with an ever-changing environment.

Stakeholders:

- **Cardholders** – Directly affected by monetary loss and the inconvenience of false declines on legitimate purchases
- **Banks** – Hold financial liability for confirmed fraudulent transactions and face compliance obligations as well as risk reputational damage from improper fraud handling
- **Merchants** – Soak up costs through chargebacks and loss of goods or services already sourced
- **Payment networks** – Rely heavily on system-wide trust in the network to maintain usage

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Practical Relevance:

Effective fraud detection systems reduce financial losses in payments whilst maintaining customer trust and convenience. The connection between catching fraud and avoiding false alarms is integral to designing a proper system. A model that's tuned solely for accuracy can perform misleadingly well by assuming legitimate for almost every transaction, given how rare fraud is in the data. This'll grab more attention to evaluation strategy in the later stages of the project and shows an actual industry relevant trade off that data driven fraud systems must navigate in practice.

Task 2: Data Acquisition, Inspection, and Documentation

Yaacoub: The dataset contains of 284,807 records across 31 columns. All columns are numeric: 30 are float64 (Time, V1-V28, Amount), and Class is int64, using binary values of 0 (legitimate) and 1 (fraudulent). Time is elapsed seconds since the first transaction (0 to 172,792, roughly 48 hours), not timestamps. V1-V28 are anonymised, PCA-transformed features. Their original variables were not disclosed for security, so no domain level interpretation is possible, and their near-zero means reflect PCA's zero centring property. Amount ranges from $0 to $25,691.16, with a mean of $88.35 but a standard deviation of $250.12, indicating a right-skewed distribution. Inspection found no missing values across any columns. 1,081 duplicate rows were found. Unique value counts supported this: Time showed 124,592 unique values, V1-V28 showed an identical 275,663, and Amount showed 32,767. The identical count across all 28 PCA components suggested duplicates spanned rows rather than individual features. Class distribution showed class imbalance: 284,315 legitimate transactions (99.83%) versus 492 fraudulent (0.17%). No negative amounts were found. Two records have Time = 0, consistent with being the dataset's first observations rather than invalid data. No inconsistent entries or formatting issues, given the dataset is numeric with no text or date fields.

Key Assumptions: Time is treated as elapsed seconds only, not time of day, unless done later. V1-V28 are treated as numerical predictors with no real world meaning. The 1,081 duplicates are assumed redundant rather than repeat transactions, given no ID field exists to confirm otherwise. This will be touched on in Task 3. Records with Time = 0 are treated as valid.

Matthew: pandas was used to load the dataset in python, it showed 284,807 records from 31 columns. 30 columns were float64 (they were: time, V1-V28, and Amount) and the class being targeted was in int64. Time records the amount of time passed in seconds since the first transaction (from 0 to 172,792 seconds which is around 48 hours), which aren't real time stamps. V1-V28 are PCA components that were anonymised with source variables that are not disclosed; therefore, they cannot be assigned domain-level meaning to them individually. Amount has a range from $0 to 25,691.16, with a mean of $88.35 and a standard deviation of$250.12, meaning this is Skewed heavily to the right. There were no missing values that were found across any of the columns, but a total of 1,081 duplicate rows were found. Value counts that are unique backed this up. 124,592 values were found unique in Time, V1-V28 had an identical 275,663 and Amount had 32,767, PCA counts that matched suggests that duplication goes over full rows rather than isolated feature overlaps. Heavy Imbalance was confirmed by class distribution: 284,315 transactions were legitimate (99.83%) while 492 were fraudulent (0.17%). No values from Amount or Time were found to be negative; two records were found to sit at Time=0, consistent with the earliest transactions of the dataset rather than an error. No issues exist with the formatting or entries because every column is numeric.

Key assumptions: Time can't be converted to a real clock's time, since only seconds passed from a random starting point is measured with no actual time of day given. V1-V28 are treated as unclear numeric predictors because no real-world meaning can be assigned to them and the source variables of their PCA were withheld. The duplicate rows are assumed as data artifacts rather than actual repeating transactions; with 28 of the PCA columns being continuous, two transactions that aren't related matching by coincidence through all of them is implausible. The 2 records with Time = 0 are treated as valid because a value of 0 is to be expected for the transaction which the dataset's clock started to count from; therefore, it isn't a data error.

Task 3: Data Cleansing and Transformation

Yaacoub: Of the checks from Task 2, missing values, invalid values, inconsistent categories, and formatting issues needed no action, since none were found. The 1,081 duplicate rows were identified earlier and addressed and by examining their class distribution, it showed 32 fraud cases and 1,790 legitimate cases among them. Since no transaction ID exists to confirm them, duplicates were removed to avoid data leakage between training and test data, reducing the dataset from 284,807 to 283,726 rows. Outliers in Amount were checked using the IQR method, which flagged 31,685 transactions (11.17%) as outliers. Given the high amount, this was taken as a sign the method doesn't work on Amount's right-skewed distribution, rather than evidence of invalid data. Checking the class distribution within these outliers showed 87 as fraudulent (0.275%) versus the datasets fraud rate of 0.17%, meaning fraud is slightly overrepresented among high value transactions. Given this, outliers were retained rather than removed. Instead, a log transformation (log1p) was used on Amount to address the right-skew, reducing skewness from 16.979 to 0.161 without removing data. Time and the log-transformed Amount were also standardised using StandardScaler, bringing both onto a comparable scale to the standardised PCA components (V1-V28). These cleaning and transformation steps were made into a reusable function, clean_data(), which can reproduce the manually cleaned dataset when used on an untouched raw copy of the data. The final cleaned dataset has 283,726 rows and 31 columns.

Matthew: None of the checks from Task 2(missing values, invalid values, inconsistent categories, formatting) required action as there were none to report. The 1,081deplicates were handled by drop_duplicates(keeep=''first), reducing the size of the dataset to 283,726 rows from 284,807. The main problem with simply including the duplicates is with the train/test split; a real risk of duplicating a transaction into both sets and model will appear highly accurate on "test" set as essentially, it's repeating an answer it's already seen.

Outliers in Amount were identified not by IQR but using a Z-score calculation; transactions with a Z-score > 3 standard deviations from the mean. This identified 4,063 transactions (1.43% of the dataset which is quite small for Z-scores and possibly more conservative than using an IQR-based threshold on such a highly skewed dataset), and only 11 of these (0.27%) were fraud cases, not much different form the dataset's baseline fraud rate of 0.17%, so there was no strong case for removing the high amounts. Therefore, outliers were left in, assuming they were mostly not an indication of error, but part of normal variation in spending, and that deleting them might also delete genuine fraudulent cases. Instead of removing outliers, skew in the Amount variable was addressed by Yeo-Johnson power transform which finds the best value of 'a' for the equation and handles Amount = 0 natively. Skew for Amount was reduced from 16.98 to ~0.02 (almost 0).

Time values were scaled using RobustScaler, as it is more resistant to the outlying values that were deliberately preserved, as it scales using the median and interquartile range, rather than the mean and standard deviation. The processing was put into a function clean_data() which worked well to reproduce the manual cleaning steps.

Task 4: Exploratory Data Analysis and Visualisation

Yaacoub:

Class Distribution: The bar chart of Class shows an imbalanced target variable: legitimate transactions (Class = 0) outnumber fraudulent ones (Class = 1), consistent with the 99.83% vs 0.17% split shown in Task 2. On a linear scale, the fraud bar is barely visible; only when plotted on a log scale does it become clear. This confirms that modelling approach used later must take the imbalance into account, since a model predicting legitimate for every transaction would still get 99.8% accuracy whilst being practically useless.

Feature Distribution: The Time_scaled histogram isn't uniform. It shows peaks and dips across the range, showing the 48-hour span of the dataset, where transaction volume rises and falls with time of day. The Amount_scaled histogram is concentrated toward the lower range (approx. -2 to 1), then falls off past 1. The expected shape for transaction amounts even after log-transformation and scaling, with purchases being small to moderate and less as the amount increases.

Transaction Amount by Class: The boxplot of Amount_scaled by Class shows a difference in distribution shape between the two classes. Legitimate transactions have a median close to 0 with a tighter IQR (approx. -0.76 to 0.73), but a long tail of high value outliers reaching up to 4.22. Fraudulent transactions show a lower median (approx. - 0.47) but a wider IQR (approx. -1.48 to 0.92) and a lower maximum (2.72), with no extreme outliers past the whiskers. Fraud does not follow the same "mostly small, sometimes very large" pattern in legitimate spending.

Correlation Analysis: A correlation heatmap confirmed V1-V28 are mostly unrelated with one another, consistent with their PCA origin. Against Class, the strongest negative correlations were V17 (-0.313), V14 (-0.293), and V10 (-0.207), while the strongest positive correlation was V11 (0.149), followed by V4 (0.129) and V2 (0.085). Amount_scaled (-0.008) and Time_scaled (-0.012) both showed weak correlation with Class, suggesting they are weak linear predictors of fraud on their own.

Missing Value Check: A visual missing value heatmap of the final dataset confirmed no missing values across any feature, consistent with Task 2 findings of zero nulls across 31 columns.

Key Data Insights: This analysis showed findings that shape how the dataset should be modelled going forward. Severe class imbalance is a huge factor, with only 473 of 283,726 transactions (0.167%) fraudulent, needing evaluation metrics past accuracy and imbalance handling techniques. Time_scaled's peak and dip pattern reflects daily transaction cycle and might have predictive value, despite showing negligible correlation with fraud. Amount_scaled stays right-skewed even after transformation, consistent with spending behaviour, and fraud and legitimate transactions differ in amount distribution shape rather than just central tendency. A subset of PCA-derived features (V17, V14, V12, V10, V11, V4) carry most of the linear signal for fraud detection, suggesting feature selection could be worthwhile at the modelling stage. Data quality stays strong throughout, with Task 3 cleaning decisions reflected here. Overall, the

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dataset is suited to fraud classification, though the class imbalance and low interpretability of PCA-derived features are challenges to address in later stages.

Independent-Samples T-Test: Transaction Amount by Class:

Null hypothesis: there is no difference in mean transaction amount between legitimate and fraudulent transactions.

Alternative hypothesis: there is a difference. Significance level set at a = 0.05. Welch's t-test comparing Amount_scaled between legitimate (n = 283,253, mean = 0.0003) and fraudulent (n = 473, mean = -0.1908) transactions returned t = 3.11, p = 0.00197. Since p is below 0.05, the null hypothesis is rejected. There is significant evidence the mean transaction amount differs by class. However, given the sample size, this significance does not mean a large difference, a point also shown in the weak correlation (-0.008) found between Amount_scaled and Class.

Chi-Square Test of Independence: Transaction Amount Category vs Class:

Since the dataset has no categorical variables, Amount_scaled was put into four quartile categories (Low, Medium, High, Very High).

Null hypothesis: fraud rate is independent of amount category

Alternative hypothesis: fraud rate depends on amount category. Significance level: a = 0.05. The contingency table showed 213 and 168 fraud cases in Low and Very High bins, versus only 47 and 45 in Medium and High bins, despite almost identical total transactions per bin (70,000-71,000). This gave us x^2=185.42, p < 0.0001, df = 3, rejecting the null hypothesis. Fraud rate is associated with amount category, concentrated at both extremes and not the middle range. This indicates association and not causation, and the bin boundaries were a modelling choice rather than grouping, so different binning could show different category-level patterns.

Matthew: The Class count bar chart with the exact numbers on top of each bar displayed the imbalance in numbers: 283,253 geunine transaction and 473 fraud, 0.17% chance of being a fraud.

Examining the two engineered features, the KDE plot for Time_*scaled oscillates repeatedly over the domain as expected given the natural fluctuations of transactions over the 48-hour span of the data rather than a constant rate. Amount*_transformed was enarly normally distributed with a mean of 0 and a standard deviation of one. This distribution profile is expected, as Yeo-Johnson transform standardises its output by design.

Visualistation using a violin plot demonstrated a discernible gap separating Amount_transformed into two groups based upon their assigned class attribute: values

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corresponding to fraud events tended to have lower distributions (median = -0.44) in comparison with values of legitimate transactions (median = 0.03). Furthermore, there was a significant overlap between these distributions, and they displayed wider and more shallow spread rather than sharply defined peaks. In layman's terms, fraud cases tended to execute transactions with lower amounts than legitimate consumers. However, due to overlap between the two groups, neither Amount alone was able to sufficiently classify transactions.

Focusing on correlation, analysis was confined to the 10 features exhibiting the greatest predictive potential related to Class via heatmap: the top three predictor values were V17, V14 and V12 (with corresponding correlation coefficients of -0.31, -0.29 and – 0.25). V10 and V16 were the following highest predictors. Interestingly, neither of the engineered variables (Time_*scaled and Amount*_Transformed) had considerable predictive value correlating closely to Class target variable (with both yielding values just under -0.01). The primary driver in the correlation with Class was thus driven by the PCA components.

The bar chart for missing values on all 31 columns of the dataset was empty as there were no missing values in all columns.

Summary and Implications: Given the observed severe class imbalance, metrics like accuracy alone would provide an inaccurate representation of model performance; therefore, recall, precision, PR-AUC scores, and anomaly detection-specific modelling (i.e. over/under sampling) will be paramount. The time-related nature depicted by the cyclic nature of the KDE plot for Time_*scaled may suggest that extracting time-based features (as opposed to simply measuring time elapsed) may provide greater signal in future models. The Yeo-Johnson transformation was effective with Amount_transformed being transformed to a relatively normal distribution and removing any existing skew that could complicate modelling. The linear correlations revealed a concentrating signal into a handful of PCA components rather than a diffuse spread among all predictor features; consequently, reducing the set of predictors to be utilize during modelling appears logical rather than inputting all 30 predictors equally.*

Independent-Samples T-Test: Transaction Amount by Class

The question here is simple: does the average amount per transaction in fact differ between fraud and legitimate entities, or could any gap detected simply be noise? Running raw dollar values (fraud = 473 transactions, avg. Amount = $123.87, std. Dev. = $260.21, honest = 283,253 transactions, avg. Amount = $88.41, std. Dev. = $250.38), a Welch's T-test result of t = 2.96, p = 0.0032 rejected the null hypothesis that the difference in the amounts was random for a = 0.05 in favour of an alternate hypothesis where there is a difference between the sample means.

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Caution should be exercised knowing exactly what these results mean: 283,726 observations is quite a large sample size, to the effect that even a moderately small gap between 2 samples may become statistically significant. While the difference in dollar amounts, at $35, is real, it alone does not make for a good classifier (which further solidifies the correlation of Amount_transformed with Class being very weak at (-0.009) from the correlation analysis earlier and explains the heavily overlapping data when looking at the violin plots).

Chi-Square Test of Independence: Transaction Amount Category vs Class

A chi-square Test was performed for which Amount first had to be categorized into four groups based on fixed dollar thresholds (greater than or equal to $10, $10-$100, $100-$500, > $500) instead of by quartiles so that each bin represents an explicit Spending amount rang, instead of an equal percentage of the samples. After testing the null hypothesis H0 that fraud rates are not associated with Amount bin vs G1 being the case in which it does at $\alpha=0.05$: observed counts in the least spent bin and the most spent bin were 238 and 34, respectively, while 110 and 91 in the mid-left and mid-right bins respectively, despite bin membership falling within tighter rang of 100,000 (bins lowest left and highest right) up to 127,000 (bin mid-left) amount members to be more exact. This gives X^2 = 105.65 (df=3, p-value of 9.48e-10) which comfortably rejected the null hypothesis and further analysis.

Mapping back to a % fraud rate to gain more visual pattern from all four bins: as Amount bins move from least to most Amount, fraud rates decreased from 0.24% down to0.09%, then increased and went to 0.37%. In essence, data seems to be U-shaped with fraud cases clustering at either end as opposed to proportionally increasing in cost. Like previous tests, however, this result simply indicates association rather than causation with differing methods of discretization perhaps affecting location where it may become less pronounce and is dependent upon discrete bin cut points rather than an inherent characteristic of data itself.